IMPE LaTeX System

YANG Sikai

1.1.0

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XSR 1.3 . . . 9
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IMPE LaTeX System [illegible] XeLaTeX [illegible]

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XeLaTeX [illegible] IMPE [illegible] Unicode [illegible] OpenType [illegible]fontspec [illegible] XeTeX

[illegible] LaTeX [illegible] IMPE [illegible] XeLaTeX [illegible]

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[illegible] IMPE [illegible] PDF[illegible] v1.1.0 [illegible] B[illegible]

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IMPE 1.1.0 [illegible] `Full` [illegible] `Core` [illegible] XeLaTeX [illegible]

`Full`[illegible] XSR[illegible]

`Core`[illegible] XSR [illegible] TeX Live[illegible]

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[illegible] TEXMF [illegible] `TEXMFHOME/tex/latex/impe/`[illegible] `impe.sty`[illegible]`impeart.cls`[illegible] `impereport.cls`[illegible]`impebook.cls` [illegible] `impebeamer.cls`[illegible] `_zh`[illegible]`next*` [illegible]

```
# Windows
.\install.bat
# Windows
.\scripts\install.ps1

# Unix
sh ./install.sh
# Unix
sh ./scripts/install.sh
```

Windows [illegible] `-TexmfRoot` [illegible] `-SkipXsrPython`[illegible]Unix [illegible] TEXMF [illegible] TeX [illegible] `mktexlsr`[illegible] TeX [illegible]

[illegible] XSR [illegible] Python [illegible]

IMPE [illegible] XSR **1.3**[illegible] XSR [illegible] Python [illegible] Python [illegible] Python **3.11** [illegible] `venv` [illegible] `pip`[illegible]

[illegible] TeX [illegible] Python [illegible]

```
# Windows
.\scripts\install.ps1 -SkipXsrPython
```

```
# Unix
IMPE_SKIP_XSR_PYTHON=1 sh ./scripts/install.sh
```

XSR XeLaTeX Shell Escape XSR TTF OTF Python

IMPE TeX IMPE assets/fonts/ impe.local.tex

```
\SetCatalogFontRoot{D:/Fonts/IMPE}
```

IMPE XSR

Libertinus

```
\documentclass{impeart}
\UseFont{libertinus}
\begin{document}
IMPE    \LIB{Libertinus    }
\end{document}
```

XeLaTeX XSR

IMPE IMPE LaTeX

```
\documentclass[11pt,lang=zh]{impeart}

\UseFont{libertinus}
\UseFeatures{math,hyperlinks}

\title{   }
\subtitle{  IMPE  }
\author{  }

\begin{document}

\maketitle
```

```
\section{ }
    IMPE   \LIB{Hello, world.}

\end{document}
```

CMU `\UseFont{libertinus}` Libertinus `\LIB` `\UseFeatures` LaTeX

```
\documentclass{impeart}

\UseTemplateSet{
  fonts = {libertinus},
  features = {math,hyperlinks}
}
```

IMPE LaTeX

```
\documentclass{article}
\usepackage{impe}

\UseTemplateSet{
  layout = en_doc,
  fonts = {libertinus},
  features = {hyperlinks}
}
```

IMPE

`templates/` `tests/`

`article_en/` `impeart`

`article_zh/` `impeart` `lang=zh`

`book_en/` `impebook`

`book_zh/` `impebook` `lang=zh`

beamer_en/ impebeamer

beamer_zh/ impebeamer lang=zh

impe* fixture tests/

impeart impereport impebook impebeamer lang=en lang=zh

```
%
\documentclass[12pt,twoside]{impeart}

%
\documentclass[lang=zh,12pt]{impeart}

%
\documentclass[lang=zh]{impebook}

%
\documentclass[lang=zh]{impebeamer}
```

article report book beamer en_doc en_report en_book beamer zh_doc zh_report zh_book beamer_zh CMU

12pt twoside Beamer aspectratio=169 impe*_zh next*

impe.sty

```
\usepackage{impe}
```


\title \author IMPE \subtitle{...} LaTeX

```
\title{ }
\subtitle{ }
\author{ }
```

[illegible] \maketitle[illegible]

[illegible] [illegible]

[illegible] [illegible]

[illegible] \UseTemplateSet[illegible]

```
\UseTemplateSet{
  layout = <preset>,
  fonts = {<families>},
  globalfonts = {<families>},
  features = {<features>}
}
```

[illegible]

layout [illegible]

fonts [illegible] [illegible]

globalfonts [illegible]

features [illegible]

mainfonts [illegible] globalfonts [illegible] [illegible]fonts [illegible]
[illegible] [illegible]

[illegible] [illegible]

[illegible]IMPE [illegible]

```
\UseLayout{en_doc}
\UseLayouts{...}

\UseFont{libertinus}
\UseFonts{arabic,tibetan}

\UseFont{libertinus}[global]
\UseLocalFonts{arabic,tibetan} %
\UseGlobalFonts{libertinus}    %
```

```
\UseFeature{hyperlinks}
\UseFeatures{math,hyperlinks,tables}
```

[illegible]

[illegible]

IMPE [illegible] [illegible]

[illegible]

```
%       en_doc   CMU
\documentclass{impeart}
\UseTemplateSet{
  fonts = {libertinus},
  features = {math,hyperlinks}
}

%       zh_book CMU   Shanggu
\documentclass[lang=zh]{impebook}
\UseTemplateSet{
  fonts = {sanskrit,tibetan},
  features = {citations,index,hyperlinks}
}
```

[illegible] layout=[illegible]

[illegible]

IMPE [illegible] 1.1.0 [illegible]

[illegible]

[illegible] \UseFont{id} [illegible] \UseTemplateSet [illegible] fonts [illegible] IMPE [illegible] Unicode [illegible]

```
\UseFont{libertinus}
\UseFonts{sanskrit,tibetan,arabic}
\UseFont{libertinus}[local]
\UseFont{libertinus}[global]
```

\LIB Libertinus CMU [local] [global] \UseLocalFont \UseGlobalFont \UseMainFont

\SA{...} \Font{sanskrit}{...} \UseFont 59

Unicode sanskrit devanagari

1.1.0 69 85 59 Unicode chinese japanese korean

```
\UseWriting{[shanggu]chinese}
\UseWriting{[tiro]devanagari}
\UseWriting{[naskh]arabic}
```

\UseWriting{[,] } ID font= file= stylemap=native|full writing=horizontal|vertical offset= italic=zhuque

\DeclareWriting{ }{ } \Font{ }[ ]{ } \Writing{ }{ }

```
\DeclareWriting{han-note}
  {[shanggu,stylemap=full,italic=zhuque]chinese}
\begin{document}
\Font{han-note}{ }
\Font{sanskrit}{   }
\Writing{[tiro]devanagari}{   }
\end{document}
```

Unicode font= file=

stylemap=native stylemap=full docs/USEWRITING-1.0.5.md

CMU lang=zh

```
\UseGlobalFonts{cmu,libertinus}
```

[illegible]CMU [illegible] Libertinus [illegible] \UseFont{libertinus} [illegible] CMU[illegible]

[illegible] CJK [illegible] IMPE [illegible] xeCJK [illegible] $2 \times 2 \times 2$ [illegible] CJK [illegible]

[illegible] OpenType [illegible] shaping [illegible]

[illegible] Unicode [illegible] CJK [illegible]

[illegible] Unicode [illegible]

CJK [illegible] xeCJK [illegible] CJK [illegible] \JP{...}[illegible]\KR{...} [illegible]

[illegible] shaping[illegible]

[illegible] OpenType shaping[illegible] XeTeX [illegible] IMPE [illegible]

[illegible] XSR 1.3[illegible]

IMPE [illegible] XSR 1.3[illegible] glyphs [illegible]

[illegible] \UseFont{egyptian}[illegible] Unicode [illegible] \IMPEEgyptianText{...}[illegible]

[illegible] \UseFont{khitan_small} [illegible] \KHS{...} [illegible] XSR [illegible] \KHSstack [illegible] \KHSstackblock[illegible] U+16FE4 [illegible]

[illegible] \UseFeature{glyphs} [illegible] \IMPEGlyphRegister [illegible] \IMPEGlyph [illegible]

[illegible] Python [illegible] XeLaTeX Shell Escape [illegible] XSR [illegible] XSR [illegible] PDF [illegible] /ActualText [illegible] Unicode [illegible] PDF [illegible]

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[illegible] docs/FONTS.md [illegible] docs/USEWRITING-1.0.5.md[illegible]

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[illegible] IMPE [illegible] LaTeX [illegible]

en_doc[illegible]**zh_doc** A4 [illegible] 29 mm[illegible] 28 mm[illegible] 32 mm[illegible]

en_report□zh_report □□□ A4 □□□□□□□ 31 mm□□ 29 mm□ □ 29 mm□□ 34 mm□□□□□□□□

en_book□zh_book ISO B5□176×250 mm□□□□□□□□ □□ 19 mm□□□ 31 mm□□□□□ 3 mm□

beamer□beamer_zh Madrid □□□ 4:3 □□□□□

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```
\documentclass{impereport}
\geometry{a4paper,left=30mm,right=30mm}
\fancyhead[R]{  }
```

IMPE □□□ `\LayoutSetup` □ `\LayoutSet` □□□□□□□□□□□ `LayoutScope` □□□□□□□□□□□□□□□□□□

```
\documentclass[lang=zh]{impeart}
\LayoutSetup{page.inner=25mm,page.top=24mm}
\begin{document}
\begin{LayoutScope}{text.par-indent=2em}

\end{LayoutScope}
\end{document}
```

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hyperlinks □□□□PDF □□□□□□□□□□□□

citations □□ `biblatex`□`biber` □□□□□□□□□□

index □□ `imakeidx`□`xindy` □□□□□□□

tables □□□booktabs□ `longtable` □ `tabularx`□

image □□□□□□□□□□

drawing TikZ□PGF□`pgfplots` □□□□□ 1.18□□ `forest`□

glyphs □ XSR □□□□□□□□□□□□□□

headers □□ fancyhdr □□□□□

```
\UseFeatures{math,hyperlinks,tables,image,drawing,glyphs}
```

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math □□ theorem□lemma□ proposition□corollary□ definition□example□ remark □□□□□□□ □□□□□□□□□□□□□□□ Computer Modern□ □□□ Libertinus Math□□□□□□□□□□□□

```
\UseMathFont{libertinus}
\UseFeature{math}
```

citations □□□□ biblatex□ □□□□□□□ \addbibresource□ \textcite□\parencite□ \printbibliography □□□□ \UseCitationStyle □□□□ APA□GB/T 7714□ □□□□□□—□□□□□□□□□□

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```
\UseFeature{index}
\begin{document}
 \index{ }  \index{ ! }
     \index{ }
\printindex
\end{document}
```

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□□ \index[people]{...} □ \printindex[people]□□□□□□□□□□□□□□□ IMPE □□□□□ □□□□□□□□□□□□□□□ see□seealso□□□□□□ □□□□□□□□□□□□□□□□□□□□□□

IMPE □□□□□□□□□□□□□□□□□□□

```
texindy -L english -C utf8 -M <job>-impe <job>.idx
```

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□□□□□□ table□tabular□ longtable□booktabs □□□□□□□□□ □□□□ figure□\includegraphics □□□□□□□ □□□□□ tikzpicture□axis□ forest□□□□□□□□□□□□□□□□□□□□□□□□□□□

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```
\UseFeature{glyphs}
\IMPEGlyphRegister[logical={   },description={ }]
  {bs-042}{images/bs-042.svg}
\begin{document}
  \IMPEGlyph{bs-042}
\IMPEGlyph[scale=1.2,raise=1pt,trim={1pt 0pt 1pt 0pt}]{bs-042}
\end{document}
```

[illegible] logical [illegible] description [illegible] trim [illegible]

[illegible]

headers [illegible] \title [illegible] \HeaderTitle{...} [illegible] fancyhdr[illegible]

hyperlinks [illegible] hyperref [illegible] bookmark[illegible] Unicode [illegible] PDF [illegible] \label[illegible]\ref[illegible] \href[illegible]\url [illegible] docs/FEATURES.md[illegible]

[illegible]

[illegible] impe.sty[illegible] impeart[illegible]impereport[illegible] impebook[illegible]impebeamer[illegible] lang=zh [illegible] impeart_zh[illegible]impereport_zh[illegible] impebook_zh[illegible]impebeamer_zh [illegible] lang=zh[illegible] next* [illegible] nextsystem.sty [illegible]

[illegible] \UseFont [illegible] [local][illegible][global] [illegible]

[illegible] \index [illegible] \Term [illegible] \IndexTitle [illegible] \index [illegible] glossaries-extra[illegible]

[illegible] TableInlineFit[illegible]TableLong[illegible] TableBook[illegible]TableBookX[illegible] TableBookNotes[illegible]NiceBooktable*[illegible] OneImage[illegible]OneImageInline[illegible] PanelFigure[illegible]\Panel [illegible] LaTeX [illegible] ExampleBlock [illegible]

[illegible] 1.0.5 [illegible] \UseLegacyLayout[illegible]

```
\documentclass[lang=zh]{impebook}
\UseLegacyLayout
```

[illegible] report [illegible] legacy_* [illegible]

[illegible] tex/latex/nextsystem/ [illegible] 0.1.x [illegible] tex/latex/impe/[illegible] impe.local.tex[illegible]

[illegible]

[illegible]

```
package/
core/
```

catalog/

modules/

assets/

scripts/

manual/

docs/

core/

OpenType

catalog/

fontspec shaping

modules/

OpenType shaping

IMPE

docs/SYSTEM-zh.md docs/FONTS-zh.md docs/LAYOUTS-zh.md docs/FEATURES-zh.md

IMPE

docs/SYSTEM-zh.md

docs/FONTS-zh.md shaping

docs/LAYOUTS-zh.md

docs/FEATURES-zh.md

CHANGELOG.md

github.com/KelvinYangBK67/IMPE-LaTeX-System

[illegible] GitHub Issues [illegible]

github.com/KelvinYangBK67/IMPE-LaTeX-System/issues

[illegible] TEX [illegible] TEX Live [illegible] IMPE [illegible]IMPE [illegible]

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A [illegible]

[illegible] 1.1.0 [illegible]

[illegible]

[illegible] lang=en[illegible] lang=zh[illegible] \documentclass[lang=zh]{impeart} [illegible]

[illegible]	[illegible]	**lang=zh** [illegible]
impeart	en_doc[illegible]A4[illegible]CMU	zh_doc[illegible]A4[illegible]CMU [illegible]
impereport	en_report[illegible]A4[illegible]CMU	zh_report[illegible]A4[illegible]CMU [illegible]
impebook	en_book[illegible]B5[illegible]CMU	zh_book[illegible]B5[illegible]CMU [illegible]
impebeamer	beamer[illegible]Madrid 4:3[illegible]CMU	beamer_zh[illegible]Madrid 4:3[illegible]CMU [illegible]

[illegible] \usepackage{impe}[illegible]

[illegible]

[illegible]	[illegible]
\UseTemplateSet	[illegible]
\UseFont[illegible]\UseFonts	[illegible] [local] [illegible] [global][illegible]
\UseLocalFont[illegible]\UseLocalFonts	[illegible]
\UseGlobalFont[illegible]\UseGlobalFonts	[illegible]
\UseMainFont[illegible]\UseMainFonts	[illegible]
\UseLayout[illegible]\UseLayouts	[illegible]
\UseFeature[illegible]\UseFeatures	[illegible]
\SetCatalogFontRoot	[illegible]

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\subtitle	□□□□□□

\UseTemplateSet □□□□□□□ layout□fonts□ globalfonts□features□ mainfonts □□□□□□□□□□ □□□□□□□□□□□□□□□□□□□□□ □□□□□□□□□□□□□□□□□□□□□□□□

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```
\UseWriting{[shanggu]chinese}
\DeclareWriting{han-note}{[shanggu,italic=zhuque]chinese}
\Font{han-note}{ }
\Writing{[tiro]devanagari}{   }
```

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\UseWriting	□□□□□□□□□□□□□□□□□□
\DeclareWriting	□□□□□□□□□□
\Font	□□□□□□□□□□□□□□□
\Writing	□□□□□□□□□□□□□□

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en_doc□zh_doc	A4 □□□□□
en_report□zh_report	□□□ A4 □□□□□
en_book□zh_book	ISO B5 □□□□□
beamer□beamer_zh	Madrid 4:3 □□□
beamer_clean□beamer_clean_zh	□□□□□□□□□□
\LayoutSetup□\LayoutSet	□□□□□□□□
LayoutScope	□□□□□□□□□□
\HeaderStyle□\HeaderTitle	□□□□□□□□□□□

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`math`	AMS □□□□□□□□□□ `\UseMathFont`□
`hyperlinks`	`hyperref`□`bookmark`□□□□□□
`citations`	`biblatex` □□□□□□□□
`index`	`\makeindex`□`\index`□`\printindex`□
`tables`	`tabular`□`booktabs`□`longtable`□
`image`	`\includegraphics` □□□□
`drawing`	TikZ□`axis`□`forest`□
`glyphs`	`\IMPEGlyphRegister`□`\IMPEGlyph`□
`headers`	□□ `fancyhdr` □□□□

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`noto`	L+G	`\NOT`	□□ Noto □□□□
`times`	L+G	`\TIM`	Times□Arial□Consolas □□□□
`gentium`	L+G	`\GEN`	Gentium Plus
`charis`	L+G	`\CHA`	Charis SIL
`libertinus`	L+G	`\LIB`	Libertinus □□□□□□□□□□□
`mlmodern`	L	`\MLM`	MLModern □□□□
`anatolian`	L	`\CA`	□□□□
`coptic`	L	`\CO`	□□□□
`bopomofo`	L	`\ZY`	□□□□
`cuneiform`	L	`\CU`	□□□□
`glagolitic`	L	`\GL`	□□□□□
`italic`	L	`\OI`	□□□□□
`hungarian`	L	`\OH`	□□□□□
`runic`	L	`\RU`	□□□
`armenian`	L	`\HY`	□□□□□
`georgian`	L	`\KA`	□□□□□

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hindi	L+R	\HI	□□□□□□□□□□
sanskrit	L+R	\SA	□□□□□□□□□□□□□□□□□□□□
devanagari	L	\DEV	□□□□□
tamil	L	\TA	□□□□
brahmi	L	\BR	□□□□
tibetan	L+R	\TI	□□□□□□□□□□□□
segoe	L	\SEG	Segoe UI Historic
thai	L	\TH	□□
arabic	L	\AR	□□□□□□□□□□□
urdu	L	\UR	□□□□□□□□□
aramaic	L	\IA	□□□□□
nabataean	L	\NB	□□□□
hebrew	L	\HE	□□□□
syriac	L	\SY	□□□□
syriac_eastern	L+R	\SYE	□□□□□□□□□□□□
kharosthi	L	\KH	□□□
egyptian	L	\EG	□□□□ XSR □□□□
khitan_small	L	\KHS	□□□□□ XSR □□
pahlavi_parthian	L+R	\PAR	□□□□□
pahlavi_inscriptional	L+R	\PAH	□□□□□□
pahlavi_psalter	L	\PSP	□□□□□□
avestan	L	\AV	□□□□□
manichaean	L	\MA	□□□
phoenician	L	\PH	□□□□
samaritan	L	\SM	□□□□□
sogdian	L	\SG	□□□
sogdian_old	L+R	\SGO	□□□□□□□□□□□
chinese_simplified	L+R	\SC	□□□□ CJK □□
chinese_traditional	L+R	\TC	□□□□ CJK □□
japanese	L+R	\JP	□□
wenjin	L+R	\WJ	□□□□□□□ CJK □□□□
shanggu	G	—	□□□□□□□ CJK □□
sim	G	—	□□□□□□□□
korean	L+R	\KR	□□□
tangut	L+R	\TG	□□□□□□□□ CJK □□□□
mongolian	L	\MO	□□□□□□□□□□
mongolian_baiti	L	\MOb	□□□ Baiti □□□□□□□

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manchu	L	\MC	□□□□□□□□□
turkic	L	\OT	□□□□
uyghur	L	\UY	□□□□□□□□□□□□□
vietnamese_quocngu	L	\VI	□□□□□□□□
vietnamese_hannom	L	\HN	□□□□□□□□

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basic	basic	basic
latin	basic	latin
greek	basic	greek-only
cyrillic	basic	cyrillic
armenian	armenian	armenian
georgian	georgian	georgian
hebrew	hebrew	hebrew
arabic	naskh	arabic
syriac	syriac	syriac
samaritan	samaritan	samaritan
aramaic	aramaic	imperial-aramaic
nabataean	nabataean	nabataean
phoenician	phoenician	phoenician
avestan	avestan	avestan
pahlavi-parthian	pahlavi-parthian	inscriptional-parthian
pahlavi-inscriptional	pahlavi-inscriptional	inscriptional-pahlavi
pahlavi-psalter	pahlavi-psalter	psalter-pahlavi
manichaean	manichaean	manichaean
sogdian	sogdian	sogdian
old-sogdian	old-sogdian	old-sogdian
old-uyghur	old-uyghur	old-uyghur
devanagari	devanagari	devanagari
tamil	tamil	tamil
thai	thai	thai
brahmi	brahmi	brahmi

ID		
kharoshthi	kharoshthi	kharoshthi
tibetan	tibetan	tibetan
coptic	coptic	coptic-main, coptic-legacy
glagolitic	glagolitic	glagolitic
old-italic	old-italic	old-italic
old-hungarian	old-hungarian	old-hungarian
runic	runic	runic
old-turkic	old-turkic	old-turkic
carian	carian	carian
cuneiform	cuneiform	cuneiform
egyptian	egyptian	egyptian
khitan-small	khitan-small	khitan-small
tangut	tangut	tangut
han	shanggu	han
chinese	shanggu	chinese
japanese	japanese	japanese
korean	korean	korean
kana	japanese	kana
hangul	korean	hangul
bopomofo	iming	bopomofo
mongolian	mnglwhite	mongolian
jurchen	—	jurchen
seal	—	seal
proto-cuneiform	—	proto-cuneiform
bengali	—	Bengali, BengaliSupplement
gurmukhi	—	Gurmukhi
gujarati	—	Gujarati
oriya	—	Oriya
telugu	—	Telugu
kannada	—	Kannada
malayalam	—	Malayalam
sinhala	—	Sinhala
lao	—	Lao
myanmar	—	Myanmar, MyanmarExtendedA, MyanmarExtendedB

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ethiopic	—	Ethiopic, EthiopicExtended, EthiopicExtendedA, EthiopicExtendedB
khmer	—	Khmer, KhmerSymbols
ogham	—	Ogham
cherokee	—	Cherokee, CherokeeSupplement
nko	—	NKo
tifinagh	—	Tifinagh
yi	—	YiSyllables, YiRadicals
adlam	—	Adlam
thaana	—	Thaana
mandaic	—	Mandaic

kharoshthi □□□□□□□□□□□□□□ □□□ kharosthi □□□□□□□□□□□□□□ □□□□□□□□□□□

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basic	noto
cmu	cmu
times	times
gentium	gentium
charis	charis
libertinus	libertinus
latinmodern	mlmodern
chinese-simplified	chinese_simplified
chinese-traditional	chinese_traditional
japanese	japanese
korean	korean
wenjin	wenjin
shanggu	shanggu
armenian	armenian
devanagari	devanagari
tamil	tamil
georgian	georgian

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tibetan	tibetan
hebrew	hebrew
arabic	arabic
syriac	syriac
syriac-eastern	syriac_eastern
syriac-western	syriac
thai	thai
thai-looped	thai
naskh	arabic
kufi	arabic
nastaliq	urdu
aref	arabic
davidlibre	hebrew
khawa	tibetan
carian	anatolian
coptic	coptic
cuneiform	cuneiform
glagolitic	glagolitic
old-italic	italic
old-hungarian	hungarian
runic	runic
brahmi	brahmi
kharoshthi	kharosthi
aramaic	aramaic
nabataean	nabataean
phoenician	phoenician
avestan	avestan
pahlavi-parthian	pahlavi_parthian
pahlavi-inscriptional	pahlavi_inscriptional
pahlavi-psalter	pahlavi_psalter
manichaean	manichaean
samaritan	samaritan
sogdian	sogdian
old-sogdian	sogdian_old
old-turkic	turkic
old-uyghur	uyghur
egyptian	egyptian

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khitan-small	khitan_small
tangut	tangut
mongolian	mongolian
mnglwhite	mongolian
mngltitle	mongolian
mnglwriting	mongolian
mnglart	mongolian
monbaiti	mongolian_baiti
abxy	manchu
segoeuihistoric	segoe
tiro	sanskrit
iming	bopomofo
zhuque	shanggu
simsun	sim
simhei	sim
simfang	sim
simkai	sim
arial	times
consolas	times
nomnatong	vietnamese_hannom
minhnguyen	vietnamese_hannom
gothicnguyen	vietnamese_hannom
syrcom	syriac
syrcom-edessa	syriac
syrcom-midyat	syriac
syrcom-jerusalem	syriac
syrcom-jerusalem-bold	syriac
syrcom-adiabene	syriac_eastern
syrcom-ctesiphon	syriac_eastern
tangut-n4694	tangut
new-tangut-std	tangut

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zhuque	ZhuqueFangsong-Regular.ttf
simkai	simkai.ttf
simfang	simfang.ttf
aref	ArefRuqaa-Regular.ttf
naskh	NotoNaskhArabic-Regular.ttf
nastaliq	NotoNastaliqUrdu-Regular.ttf

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impeart_zh	impeart □ lang=zh
impereport_zh	impereport □ lang=zh
impebook_zh	impebook □ lang=zh
impebeamer_zh	impebeamer □ lang=zh
nextart□nextreport□nextbook□nextbeamer	□□□□□ impe* □□
next*_zh	□□□□□□□□ lang=zh
nextsystem.sty	impe.sty

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`\IndexTitle`	□□□□□□□□□□□
`TableInlineFit`□`TableLong`	□□□□□□□
`TableBook*`□ `NiceBooktable*`	□□□□□□□□□□□
`OneImage*`□`PanelFigure*`□ `\Panel`	□□□□□□□□□□□
`ExampleBlock`	□□□□□□□

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IMPE LaTeX System

Canonical Typesetting Showcase

YANG Sikai

v1.1.0

Abstract

This document is the canonical visual specimen for the current IMPE LaTeX System public interface. It combines a local-family style audit, a deliberately small Unicode-range routing regression, dense multilingual paragraph composition with explicit local overrides, right-to-left text, vertical writing, generic complex shaping, CJK and Han family selection, and the principal document features provided by IMPE. It is intended both as a public showcase and as a release-time visual regression specimen.[1]

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[1] The showcase is intentionally broader than the focused regression fixtures under `tests/`; those fixtures isolate individual failures.

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1 Local Font Families and Styles

Table 1 pairs concise script specimens with eight corresponding text styles. Repeated specimens make differences between typefaces directly comparable; where a face is not supplied, the registered family's normal fallback applies. Egyptian format controls and Khitan clusters use their native two-dimensional layouts; Inner Asian samples use vertical writing. The matrix continues across landscape pages as needed.

2 Multilingual and Mixed-Script Composition

Written traditions carry voices across languages, centuries, and regions. The passages here bring together reflections on books and learning alongside short proverbial and scriptural lines, composed in their own writing systems.

2.1 Polyglot Composition

The same page can accommodate many scripts without assigning any single one the role of a translation or a transcription. In the following continuous paragraph, five writing systems are set directly in Unicode, while the others retain their chosen local typefaces.

Գրքերը մեզ պատմում են անցյալի մարդկանց մասին: ძველ წიგნებში მრავალი თაობის აზრი და გამოცდილებაა შემონახული. Ογοϩ ⲡⲓⲗⲟⲅⲟⲥ ⲁⲛⲱⲱⲙ, ⲛⲟⲩⲥⲁⲣⲝ ⲟⲩⲟϩ ⲁⲛⲱⲱⲙ ⲛⲏⲏⲧⲉⲛ. 𓅱𓏏 𓊪𓎛𓐍𓄿𓂋 𓇋𓂋𓏏𓈙 𓐍𓄿𓏏𓏏𓈙, 𓅓 𓐍𓄿𓏏𓏏𓈙 𓇋𓂋𓏏𓈙 𓎛𓏏 𓇋𓈙𓋴𓏏. विद्या ददाति विनयं । கற்க கசடறக் கற்பவை கற்றபின் நிற்க அதற்குத் தக. ཤེས་རབ་དང་སྦྱིང་རྗེའི་མིའི་ཚོས་ཀྱི་རྒྱ་ཡིན། الكتاب خير جليس، والعلم نور. ہماری کتابیں ہمیں ماضی کی دنیا سے ملاتی ہیں۔ החכמה טובה

Table 1: Typeface specimens in eight styles.

Typeface	Regular	Bold	Italic	Bold italic	Sans	Sans bold	Sans italic	Sans bold italic
Latin typefaces								
CMU Serif	Latina	**Latina**	*Latina*	***Latina***	Latina	**Latina**	Latina	**Latina**
Noto	Latina	**Latina**	*Latina*	***Latina***	Latina	**Latina**	*Latina*	***Latina***
Times	Latina	**Latina**	*Latina*	***Latina***	Latina	**Latina**	*Latina*	***Latina***
Gentium Plus	Latina	**Latina**	*Latina*	***Latina***	Latina	**Latina**	*Latina*	***Latina***
Charis SIL	Latina	**Latina**	*Latina*	***Latina***	Latina	**Latina**	*Latina*	***Latina***
Libertinus	Latina	**Latina**	*Latina*	***Latina***	Latina	**Latina**	*Latina*	**Latina**
MLModern	Latina	**Latina**	*Latina*	***Latina***	Latina	**Latina**	*Latina*	***Latina***
European and Caucasian scripts								
Carian	⊕ΔΑΙϑΘ	⊕ΔΑΙϑΘ	⊕ΔΑΙϑΘ	⊕ΔΑΙϑΘ	⊕ΔΑΙϑΘ	⊕ΔΑΙϑΘ	⊕ΔΑΙϑΘ	⊕ΔΑΙϑΘ
Coptic	ⲛⲟⲩⲧⲉ	ⲛⲟⲩⲧⲉ	ⲛⲟⲩⲧⲉ	ⲛⲟⲩⲧⲉ	ⲛⲟⲩⲧⲉ	ⲛⲟⲩⲧⲉ	ⲛⲟⲩⲧⲉ	ⲛⲟⲩⲧⲉ
Glagolitic	ⰶⰻⱀⱁⰿⰻ	ⰶⰻⱀⱁⰿⰻ	ⰶⰻⱀⱁⰿⰻ	ⰶⰻⱀⱁⰿⰻ	ⰶⰻⱀⱁⰿⰻ	ⰶⰻⱀⱁⰿⰻ	ⰶⰻⱀⱁⰿⰻ	ⰶⰻⱀⱁⰿⰻ
Old Italic	ᛖI ᚾV⊗I	ᛖI ᚾV⊗I	ᛖI ᚾV⊗I	ᛖI ᚾV⊗I	ᛖI ᚾV⊗I	ᛖI ᚾV⊗I	ᛖI ᚾV⊗I	ᛖI ᚾV⊗I
Old Hungarian	ᏗᎴᎷᏋᏂ	ᏗᎴᎷᏋᏂ	ᏗᎴᎷᏋᏂ	ᏗᎴᎷᏋᏂ	ᏗᎴᎷᏋᏂ	ᏗᎴᎷᏋᏂ	ᏗᎴᎷᏋᏂ	ᏗᎴᎷᏋᏂ
Runic	ᛞᚢᚦᚠᚱ᛬	ᛞᚢᚦᚠᚱ᛬	ᛞᚢᚦᚠᚱ᛬	ᛞᚢᚦᚠᚱ᛬	ᛞᚢᚦᚠᚱ᛬	ᛞᚢᚦᚠᚱ᛬	ᛞᚢᚦᚠᚱ᛬	ᛞᚢᚦᚠᚱ᛬
Armenian	հայերեն	**հայերեն**	հայերեն	**հայերեն**	հայերեն	**հայերեն**	հայերեն	**հայերեն**
Georgian	ქართული	**ქართული**	ქართული	**ქართული**	ქართული	**ქართული**	ქართული	**ქართული**
South Asian and Tibetan scripts								
Devanagari ·Hindi	देवनागरी	**देवनागरी**	देवनागरी	**देवनागरी**	देवनागरी	**देवनागरी**	देवनागरी	**देवनागरी**
Devanagari ·Sanskrit	देवनागरी	देवनागरी	देवनागरी	देवनागरी	देवनागरी	देवनागरी	देवनागरी	देवनागरी
Devanagari	देवनागरी	**देवनागरी**	देवनागरी	**देवनागरी**	देवनागरी	**देवनागरी**	देवनागरी	**देवनागरी**
Tamil	தமிழ்	**தமிழ்**	*தமிழ்*	***தமிழ்***	தமிழ்	**தமிழ்**	தமிழ்	**தமிழ்**

Typeface	Regular	Bold	Italic	Bold italic	Sans	Sans bold	Sans italic	Sans bold italic
Brahmi	𑀤𑁆𑀰	𑀤𑁆𑀰	𑀤𑁆𑀰	𑀤𑁆𑀰	𑀤𑁆𑀰	𑀤𑁆𑀰	𑀤𑁆𑀰	𑀤𑁆𑀰
Brahmi ·Segoe	𑀤𑁆𑀰	𑀤𑁆𑀰	𑀤𑁆𑀰	𑀤𑁆𑀰	𑀤𑁆𑀰	𑀤𑁆𑀰	𑀤𑁆𑀰	𑀤𑁆𑀰
Kharoshthi	𐨪𐨿	𐨪𐨿	𐨪𐨿	𐨪𐨿	𐨪𐨿	𐨪𐨿	𐨪𐨿	𐨪𐨿
Tibetan	བོད་ཡིག	**བོད་ཡིག**	ཨོྃ་ཨཱཿ	**ཨོྃ་ཨཱཿ**	བོད་ཡིག	**བོད་ཡིག**	ཨོྃ་ཨཱཿ	**ཨོྃ་ཨཱཿ**
Thai	ภาษาไทย	**ภาษาไทย**	ภาษาไทย	**ภาษาไทย**	ภาษาไทย	**ภาษาไทย**	ภาษาไทย	**ภาษาไทย**
West Asian and Iranian scripts								
Arabic	العربية	العربية	العربية	العربية	العربية	**العربية**	العربية	**العربية**
Arabic ·Nastaliq	نستعليق	نستعليق	نستعليق	نستعليق	نستعليق	نستعليق	نستعليق	نستعليق
Imperial Aramaic	𐡀𐡓𐡌𐡉𐡀	𐡀𐡓𐡌𐡉𐡀	𐡀𐡓𐡌𐡉𐡀	𐡀𐡓𐡌𐡉𐡀	𐡀𐡓𐡌𐡉𐡀	𐡀𐡓𐡌𐡉𐡀	𐡀𐡓𐡌𐡉𐡀	𐡀𐡓𐡌𐡉𐡀
Nabataean	𐢇𐢁𐢋𐢂	𐢇𐢁𐢋𐢂	𐢇𐢁𐢋𐢂	𐢇𐢁𐢋𐢂	𐢇𐢁𐢋𐢂	𐢇𐢁𐢋𐢂	𐢇𐢁𐢋𐢂	𐢇𐢁𐢋𐢂
Hebrew	עברית	**עברית**	עברית	**עברית**	עברית	**עברית**	עברית	**עברית**
Syriac	ܣܘܪܝܝܐ	**ܣܘܪܝܝܐ**	ܣܘܪܝܝܐ	ܣܘܪܝܝܐ	ܣܘܪܝܝܐ	**ܣܘܪܝܝܐ**	ܣܘܪܝܝܐ	**ܣܘܪܝܝܐ**
Syriac ·Eastern	ܣܘܪܝܝܐ	ܣܘܪܝܝܐ	ܣܘܪܝܝܐ	ܣܘܪܝܝܐ	ܣܘܪܝܝܐ	**ܣܘܪܝܝܐ**	ܣܘܪܝܝܐ	**ܣܘܪܝܝܐ**
Inscriptional Parthian	𐭠𐭥𐭫𐭲𐭠	𐭠𐭥𐭫𐭲𐭠	𐭠𐭥𐭫𐭲𐭠	𐭠𐭥𐭫𐭲𐭠	𐭠𐭥𐭫𐭲𐭠	𐭠𐭥𐭫𐭲𐭠	𐭠𐭥𐭫𐭲𐭠	𐭠𐭥𐭫𐭲𐭠
Inscriptional Pahlavi	𐭠𐭥𐭫𐭲𐭠	𐭠𐭥𐭫𐭲𐭠	𐭠𐭥𐭫𐭲𐭠	𐭠𐭥𐭫𐭲𐭠	𐭠𐭥𐭫𐭲𐭠	𐭠𐭥𐭫𐭲𐭠	𐭠𐭥𐭫𐭲𐭠	𐭠𐭥𐭫𐭲𐭠
Psalter Pahlavi	𐮀𐮅𐮉𐮂	𐮀𐮅𐮉𐮂	𐮀𐮅𐮉𐮂	𐮀𐮅𐮉𐮂	𐮀𐮅𐮉𐮂	𐮀𐮅𐮉𐮂	𐮀𐮅𐮉𐮂	𐮀𐮅𐮉𐮂
Avestan	𐬀𐬌𐬎𐬀	𐬀𐬌𐬎𐬀	𐬀𐬌𐬎𐬀	𐬀𐬌𐬎𐬀	𐬀𐬌𐬎𐬀	𐬀𐬌𐬎𐬀	𐬀𐬌𐬎𐬀	𐬀𐬌𐬎𐬀
Manichaean	𐫀𐫁𐫂𐫀	𐫀𐫁𐫂𐫀	𐫀𐫁𐫂𐫀	𐫀𐫁𐫂𐫀	𐫀𐫁𐫂𐫀	𐫀𐫁𐫂𐫀	𐫀𐫁𐫂𐫀	𐫀𐫁𐫂𐫀
Phoenician	𐤀𐤓𐤌	𐤀𐤓𐤌	𐤀𐤓𐤌	𐤀𐤓𐤌	𐤀𐤓𐤌	𐤀𐤓𐤌	𐤀𐤓𐤌	𐤀𐤓𐤌
Samaritan	ⷘⷁⷖⷭ	ⷘⷁⷖⷭ	ⷘⷁⷖⷭ	ⷘⷁⷖⷭ	ⷘⷁⷖⷭ	ⷘⷁⷖⷭ	ⷘⷁⷖⷭ	ⷘⷁⷖⷭ
Sogdian	𐼧𐼤𐼣𐼤	𐼧𐼤𐼣𐼤	𐼧𐼤𐼣𐼤	𐼧𐼤𐼣𐼤	𐼧𐼤𐼣𐼤	𐼧𐼤𐼣𐼤	𐼧𐼤𐼣𐼤	𐼧𐼤𐼣𐼤
Old Sogdian	𐽆𐽁𐽀𐽆	𐽆𐽁𐽀𐽆	𐽆𐽁𐽀𐽆	𐽆𐽁𐽀𐽆	𐽆𐽁𐽀𐽆	𐽆𐽁𐽀𐽆	𐽆𐽁𐽀𐽆	𐽆𐽁𐽀𐽆

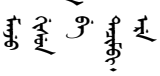
Typeface	Regular	Bold	Italic	Bold italic	Sans	Sans bold	Sans italic	Sans bold italic
Ancient Near Eastern scripts								
Cuneiform	𒀀𒂍𒄿	𒀀𒂍𒄿	𒀀𒂍𒄿	𒀀𒂍𒄿	𒀀𒂍𒄿	𒀀𒂍𒄿	𒀀𒂍𒄿	𒀀𒂍𒄿
Egyptian Hieroglyphs	𓂋𓏏𓈖𓂝	𓂋𓏏𓈖𓂝	𓂋𓏏𓈖𓂝	𓂋𓏏𓈖𓂝	𓂋𓏏𓈖𓂝	𓂋𓏏𓈖𓂝	𓂋𓏏𓈖𓂝	𓂋𓏏𓈖𓂝
East Asian scripts and Han families								
Bopomofo	ㄅㄆㄇㄈ	ㄅㄆㄇㄈ	ㄅㄆㄇㄈ	ㄅㄆㄇㄈ	ㄅㄆㄇㄈ	ㄅㄆㄇㄈ	ㄅㄆㄇㄈ	ㄅㄆㄇㄈ
Han ·Simplified	汉字	**汉字**	汉字	**汉字**	汉字	**汉字**	汉字	**汉字**
Han ·Traditional	漢字	**漢字**	漢字	**漢字**	漢字	**漢字**	漢字	**漢字**
Han ·Hiragana · Katakana	漢字かなカナ	**漢字かなカナ**	漢字かなカナ	**漢字かなカナ**	漢字かなカナ	**漢字かなカナ**	漢字かなカナ	**漢字かなカナ**
Han ·WenJin	漢字	漢字	漢字	漢字	漢字	漢字	漢字	漢字
Han ·Shanggu	漢字	**漢字**	漢字	**漢字**	**漢字**	**漢字**	**漢字**	**漢字**
Han ·Sim	汉字	汉字	汉字	汉字	汉字	汉字	汉字	汉字
Hangul ·Han	한글漢字	**한글漢字**	한글漢字	**한글漢字**	한글漢字	**한글漢字**	한글漢字	**한글漢字**
Tangut	𘜶𘝦	𘜶𘝦	𘜶𘝦	𘜶𘝦	𘜶𘝦	𘜶𘝦	𘜶𘝦	𘜶𘝦
Inner Asian scripts								
Khitan Small Script	令岑 小 分 列	令岑 小 分 列	令岑 小 分 列	令岑 小 分 列	令岑 小 分 列	令岑 小 分 列	令岑 小 分 列	令岑 小 分 列
Mongolian	ᠮᠣᠩᠭᠣᠯ	ᠮᠣᠩᠭᠣᠯ	ᠮᠣᠩᠭᠣᠯ	ᠮᠣᠩᠭᠣᠯ	ᠮᠣᠩᠭᠣᠯ	ᠮᠣᠩᠭᠣᠯ	ᠮᠣᠩᠭᠣᠯ	ᠮᠣᠩᠭᠣᠯ
Mongolian Baiti	ᠮᠣᠩᠭᠣᠯ	ᠮᠣᠩᠭᠣᠯ	ᠮᠣᠩᠭᠣᠯ	ᠮᠣᠩᠭᠣᠯ	ᠮᠣᠩᠭᠣᠯ	ᠮᠣᠩᠭᠣᠯ	ᠮᠣᠩᠭᠣᠯ	ᠮᠣᠩᠭᠣᠯ
Manchu	ᠮᡠᠨ	ᠮᡠᠨ	ᠮᡠᠨ	ᠮᡠᠨ	ᠮᡠᠨᡤᡠᠯ	ᠮᡠᠨᡤᡠᠯ	ᠮᡠᠨᡤᡠᠯ	ᠮᡠᠨᡤᡠᠯ
Old Turkic	𐰃𐰆𐰚𐰇	𐰃𐰆𐰚𐰇	𐰃𐰆𐰚𐰇	𐰃𐰆𐰚𐰇	𐰃𐰆𐰚𐰇	𐰃𐰆𐰚𐰇	𐰃𐰆𐰚𐰇	𐰃𐰆𐰚𐰇
Old Uyghur	𐽆𐽏𐽂𐽏	𐽆𐽏𐽂𐽏	𐽆𐽏𐽂𐽏	𐽆𐽏𐽂𐽏	𐽆𐽏𐽂𐽏	𐽆𐽏𐽂𐽏	𐽆𐽏𐽂𐽏	𐽆𐽏𐽂𐽏

Typeface	Regular	Bold	Italic	Bold italic	Sans	Sans bold	Sans italic	Sans bold italic
Vietnamese writing traditions								
Latin ·Quốc Ngữ	Quốc ngữ	**Quốc ngữ**	Quốc ngữ	**Quốc ngữ**	Quốc ngữ	Quốc ngữ	Quốc ngữ	Quốc ngữ
Han ·Chữ Nôm	笀喃	**笀喃**	笀喃	笀喃	**笀喃**	**笀喃**	**笀喃**	笀喃

.מכסף ומזהב .𐡀𐡋𐡋𐡄 𐡀𐡅𐡌 ,𐡌𐡏𐡃𐡓𐡀 𐡇𐡎𐡔𐡄 .𐡇𐡊𐡄 𐡆𐡍 𐡌𐡏𐡀𐡍𐡉 𐡀𐡎𐡕𐡄 𐡄𐡊 ᠣᠯᠠᠨ ᠤᠨᠰᠢᠭᠰᠠᠨ ᠬᠡᠪᠴᠢᠶᠡᠨ ᠤ ᠳᠡᠭᠡᠨ, Dǰdl. 古い図書館でカタログをめくり、昔の人の手紙を読みました。 古典을 읽으면 오늘의 질문을 다시 생각하게 됩니다. 舊書中的一句話，有時會在數百年後重新被人理解。

2.2 Mixed Writing Directions

A page may also alternate between horizontal sentences and vertically written words. Here the Mongolian and Manchu passages rise between ordinary horizontal lines, while each writing system retains its own orientation.

翻開一本舊書，紙頁上的文字把我們帶到另一個時代。 古い手紙を読むと、文字の形から書き手の暮らしまで想像できます。  오래된 기록을 읽는 일은 낯선 사람의 목소리에 귀를 기울이는 일입니다. 文字的方向雖然不同，閱讀仍讓相隔久遠的人彼此相遇。

3 Right-to-Left Scripts

Arabic, Urdu, Hebrew, and Syriac begin their lines at the right-hand edge, yet their books also accommodate dates, shelfmarks, and occasional passages in other writing systems. These pages bring together modern prose, verse, and older scriptural traditions.

3.1 Arabic

In a quiet manuscript room, a catalogue entry may be the beginning of a much longer story.

في قاعة المخطوطات رفوف من الخشب الداكن، وعلى الرفوف كتب نُسخت في مدن مختلفة وعصور متباعدة. تحمل بعض الصفحات أسماء ناسخين وقرّاء، وتحفظ الهوامش أسئلة صغيرة تركها أشخاص لا نعرف عن حياتهم إلا القليل. تُظهر إحدى النسخ آثار التصحيحات التي أضافها ناسخ لاحق، وتروي حواشيها كيف انتقل الكتاب بين أيدي القرّاء. وحين يقارن الباحث نسختين من الكتاب نفسه يجد أن اختلاف كلمة واحدة قد يكشف طريق النص من مدينة إلى أخرى. لهذا لا تكون القراءة عودة إلى الماضي وحده؛ إنها حوار جديد مع أناس حفظوا الكلمات ونقلوها إلى من جاء بعدهم.

A fully vocalized couplet by al-Mutanabbī offers another typographic rhythm, with its vowel marks and joined letterforms:

أَعَزُّ مَكَانٍ فِي الدُّنَى سَرْجُ سَابِحٍ، وَخَيْرُ جَلِيسٍ فِي الزَّمَانِ كِتَابُ.

3.2 Urdu

ایک پرانے کتب خانے میں داخل ہوتے ہی کاغذ اور روشنائی کی ایک مانوس خوشبو محسوس ہوتی ہے۔ لکڑی کی الماریوں میں رکھی ہوئی کتابوں پر زمانے کے نشان صاف دکھائی دیتے ہیں، لیکن ان کے صفحات آج بھی نئے سوال پیدا کرتے ہیں۔ بعض نسخوں کے حاشیوں میں پچھلے قارئین نے اپنے خیالات لکھے ہیں؛ کہیں کسی مشکل لفظ کا مطلب درج ہے اور کہیں صرف ایک چھوٹا سا سوالیہ نشان۔ سن ۱۹۲۷ کی ایک یادداشت میں کتاب کے مالک نے لکھا تھا کہ اسے یہ نسخہ سفر کے دوران ملا۔ اب وہی کتاب دوسرے شہر کی لائبریری میں موجود ہے، جہاں ہر نیا قاری اس کی کہانی میں اپنا حصہ شامل کرتا ہے۔ یوں ایک کتاب صرف الفاظ کا مجموعہ نہیں رہتی بلکہ مختلف زمانوں اور لوگوں کے درمیان

رابطے کا ذریعہ بن جاتی ہے۔

3.3 Hebrew

כשנכנסים לספרייה ישנה, אפשר לפעמים לשמוע את רשרוש הדפים יותר מכל קול אחר. על שולחן העץ מונחים כתבי יד, מהדורות מודפסות ומחברת של קורא שכבר איננו. בשוליו של אחד הספרים נכתבה הערה קצרה בשנת 1896, ולידה סימן שאלה קטן. כעבור דורות פותח קורא אחר את אותו העמוד וממשיך את השיחה שהתחילה לפניו. כך נשמרות בספרייה לא רק המילים שבספרים, אלא גם עקבותיהם של האנשים שקראו אותם, העירו עליהם והעבירו אותם לידיים אחרות.

The opening of Genesis presents a different Hebrew page: fully pointed consonants, cantillation marks, and the verse-ending punctuation.

בְּרֵאשִׁית בָּרָא אֱלֹהִים אֵת הַשָּׁמַיִם וְאֵת הָאָרֶץ׃

3.4 Syriac

The opening of the Gospel of John in the Syriac Peshitta places words and light at the heart of its narrative. The following continuous passage is taken from John 1:2–5.

ܗܘ ܗܘܐ ܒܪܫܝܬ ܥܡ ܐܠܗܐ ܟܠ ܫܪܝܪ ܗܘ ܡܠܬܐ ❖ ܘܡܠܬܐ ܗܘ ܗܘܬ ܠܘܬ ܐܠܗܐ ❖ ܟܠ ܡܕܝܢܬܐ ܒܗ ܐܬܒܪܝ ❖ ܘܒܚܫܘܟܐ ❖ ܢܗܝܪܐ ܒܚܫܘܟܐ ❖ ܘܚܫܘܟܐ ܠܐ ܐܕܟܝܗ

Two traditional typefaces set the same opening verse, allowing their letterforms to be read alongside one another.

General Syriac

ܗܘ ܗܘܐ ܒܪܫܝܬ ܥܡ ܐܠܗܐ ❖ ܟܠ ܡܕܝܢܬܐ ܒܗ ܐܬܒܪܝ ❖ ܘܒܚܫܘܟܐ ❖ ܢܗܝܪܐ ܒܚܫܘܟܐ ❖ ܘܚܫܘܟܐ ❖ ܠܐ ܐܕܟܝܗ

Eastern Syriac

ܗܘ ܗܘܐ ܒܪܫܝܬ ܥܡ ܐܠܗܐ ❖ ܟܠ ܡܕܝܢܬܐ ܒܗ ܐܬܒܪܝ ❖ ܘܒܚܫܘܟܐ ❖ ܢܗܝܪܐ ❖ ܘܚܫܘܟܐ ❖ ܠܐ ܐܕܟܝܗ

4 Vertical Writing

Mongolian, Manchu, and Old Uyghur are written in vertical lines. In a horizontal IMPE document, local vertical runs retain that orientation while individual word-sized units can wrap within the surrounding page. The following passages compare the two settings and their different letterforms.

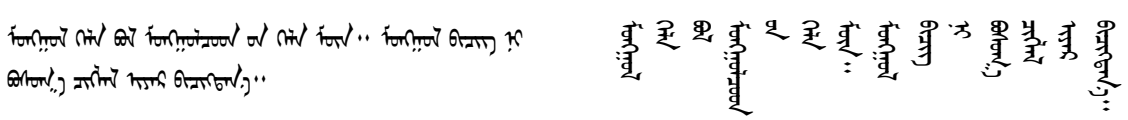
4.1 Mongolian

The same Mongolian passage appears in horizontal and vertical settings.

Horizontal reference

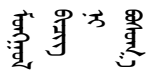
᠂ᠪᠣᠯᠣᠭᠠᠨ ᠤᠨ ᠲᠡᠮᠳᠡᠭ ᠪᠣᠯᠣᠭᠠᠨ ᠤᠨ ᠤᠨ ᠪᠣᠯᠣᠭᠠᠨ ᠂᠂ ᠪᠣᠯᠣᠭᠠᠨ ᠦᠰᠦᠭ ᠢ
ᠲᠡᠮᠳᠡᠭᠯᠡᠵᠦ ᠴᠢᠳᠠᠵᠤ ᠪᠣᠯᠣᠭᠠᠨ᠂᠂

Vertical

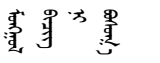


The regular Mongolian family and Mongolian Baiti give the same short phrase different appearances in vertical setting:

Mongolian



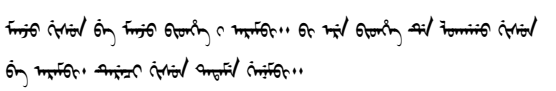
Mongolian Baiti



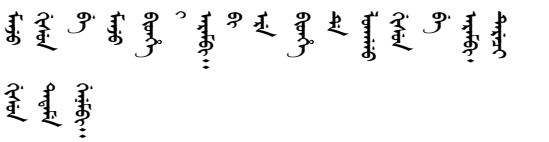
4.2 Manchu

The same Manchu passage appears in both orientations, allowing direct comparison without changing the text.

Horizontal reference



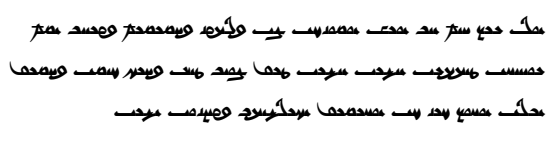
Vertical



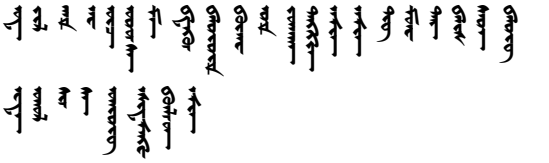
4.3 Old Uyghur

A provisional Unicode rendering based on the published transliteration of the colophon of the Turfan manuscript U 338 supplies a continuous Old Uyghur passage. It is presented as a typographic specimen rather than a critical edition.

Horizontal reference



Vertical



5 Complex Shaping and Stacked Sequences

Two kinds of composition meet on these pages. Devanagari, Tibetan, Brahmi, Kharoshthi, and Thai rely on the shaping of ordinary Unicode text by the selected OpenType fonts. Egyptian Hieroglyphs and Khitan Small Script instead use XSR to arrange signs into two-dimensional units while preserving their original Unicode sequences.

5.1 Devanagari

Devanagari makes conjunct consonants, vowel signs, and the virama visible in the shapes of its syllables. The Sanskrit family sets the forms in Table 2 with the typeface used in the Sanskrit verse.

The first three verses of the *Bhagavadgītā* (1.1–3) give a longer, continuous Sanskrit specimen with conjuncts and vowels.

> धृतराष्ट्र उवाच । धर्मक्षेत्रे कुरुक्षेत्रे समवेता युयुत्सवः । मामकाः पाण्डवाश्चैव किमकुर्वत सञ्जय ॥१ ॥ सञ्जय उवाच । दृष्ट्वा तु पाण्डवानीकं व्यूढं दुर्योधनस्तदा । आचार्यमुपसङ्गम्य राजा वचनमब्रवीत् ॥२ ॥ पश्यैतां पाण्डुपुत्राणामाचार्य महतीं चमूम् । व्यूढां द्रुपदपुत्रेण तव शिष्येण धीमता ॥३ ॥

5.2 Tibetan

Tibetan syllables can contain stacked consonants within a single *tsheg*-delimited unit. Table 3 compares short forms; the traditional Four Immeasurables prayer provides a longer passage.

སེམས་ཅན་ཐམས་ཅད་བདེ་བ་དང་བདེ་བའི་རྒྱུ་དང་ལྡན་པར་གྱུར་ཅིག ། སེམས་ཅན་ཐམས་ཅད་སྡུག་བསྔལ་དང་སྡུག་བསྔལ་གྱི་རྒྱུ་དང་བྲལ་བར་གྱུར་ཅིག ། སེམས་ཅན་ཐམས་ཅད་སྡུག་བསྔལ་མེད་པའི་བདེ་བ་དང་མི་འབྲལ་བར་གྱུར་ཅིག ། སེམས་ཅན་ཐམས་ཅད་ཉེ་རིང་ཚགས་སྡང་དང་བྲལ་བའི་བཀྲང་སྙོམ་ལ་གནས་པར་གྱུར་ཅིག།

Table 2: Devanagari shaping specimens

Sequence	Specimen
kṣa	क्ष
jña	ज्ञ
śra	श्र
stra	स्त्र
ddhya	द्ध्य
dharma	धर्म
dharmakṣetre	धर्मक्षेत्रे
saṃskṛtam	संस्कृतम्
oṃ	ॐ

Table 3: Tibetan consonant-stack specimens

Transliteration	Specimen
bsgrubs	བསྒྲུབས་
brgyud	བརྒྱུད་
bstan	བསྟན་
rgyal	རྒྱལ་
dbyangs	དབྱངས་

Sanskrit transcribed in Tibetan letters. The *Heart Sūtra* mantra illustrates Sanskrit consonant stacks, long-vowel marks, anusvāra, and the *svāhā* cluster without switching away from the Tibetan-script font:

ཏྒྱཱ་ག་ཏེ་ག་ཏེ་པཱ་ར་ག་ཏེ། པཱ་ར་སཾ་ག་ཏེ། བོ་དྷི་སྭཱ་ཧཱ།

5.3 Brahmi, Kharoshthi, and Thai

The complete Rummindei (Lumbini) pillar edict of Aśoka, in Prakrit, provides a genuine Brāhmī inscription. The same Unicode transcription is set in two regular fonts; the original inscription's line breaks are joined into one flowing paragraph without changing its characters.

Brāhmī — Noto Sans Brahmi

𑀦𑀺𑀬𑀢𑁆𑀢 𑀓𑀮𑀸𑀓𑁂 𑀫𑀳𑀸𑀬𑀸𑀦𑀲𑁆𑀫𑀺𑀁 𑀯𑀘𑁆𑀘𑀺𑀦𑀻 𑀦𑀸𑀫 𑀭𑀸𑀚𑀓𑀼𑀫𑀸𑀭 𑀓𑀸𑀮𑀸𑀪𑀺𑀲𑀺𑀢𑁂𑀦 𑀯𑁂𑀤𑀸𑀪𑀺𑀕𑀫𑀦𑀸𑀦𑀼𑀪𑀯𑁂𑀦 𑀲𑀫𑁆𑀫𑀸𑀲𑀫𑁆𑀩𑀼𑀥𑁄 𑀮𑀼𑀁𑀩𑀺𑀦𑀻 𑀕𑀸𑀫𑀺𑀦𑀻 𑀉𑀩𑀮𑀺𑀓𑀸 𑀓𑀭𑀸𑀪𑀸𑀓𑀸 𑀲𑀺𑀮𑀸 𑀣𑀽𑀪𑁄 𑀓𑀢𑁄

Brāhmī — Segoe UI Historic

𑀦𑀺𑀬𑀢𑁆𑀢 𑀓𑀮𑀸𑀓𑁂 𑀫𑀳𑀸𑀬𑀸𑀦𑀲𑁆𑀫𑀺𑀁 𑀯𑀘𑁆𑀘𑀺𑀦𑀻 𑀦𑀸𑀫 𑀭𑀸𑀚𑀓𑀼𑀫𑀸𑀭 𑀓𑀸𑀮𑀸𑀪𑀺𑀲𑀺𑀢 𑀫𑁂𑀤𑀸𑀪𑀺𑀕𑀫𑀦𑀸𑀦𑀼𑀪𑀯𑁂𑀦 𑀲𑀫𑁆𑀫𑀸𑀲𑀫𑁆𑀩𑀼𑀥𑁄 𑀮𑀼𑀁𑀩𑀺𑀦𑀻 𑀕𑀸𑀫𑀺𑀦𑀻 𑀉𑀩𑀮𑀺𑀓𑀸 𑀓𑀭𑀸𑀪𑀸𑀓𑀸 𑀲𑀺𑀮𑀸 𑀣𑀽𑀪𑁄 𑀓𑀢𑁄

Source: Aśoka's Rummindei pillar edict, based on E. Hultzsch; Unicode transcription at Brahmi script: Birthplace of the historical Buddha. Segoe UI Historic is a separately installed Microsoft font.

Kharoshthi — illustrative script transcription

𐨑𐨪𐨁𐨐𐨪𐨁 𐨑𐨁𐨐𐨪𐨁 𐨑𐨁𐨐𐨪𐨁

This short Kharoshthi passage is a modern script conversion of Sanskrit, not an attested ancient Gāndhārī inscription. Its right-to-left shaping is the focus of this specimen.

Thai — continuous contemporary prose

เมื่อเปิดหนังสือเก่าในห้องสมุด เราอาจพบลายมือของผู้อ่านจากหลายยุคสมัย บางคนขีดเส้นใต้ข้อความที่ชอบ บางคนเขียนคำถามไว้ริมหน้า และบางคนบันทึกวันเวลาที่อ่านจบ เรื่องราวเหล่านี้ทำให้หนังสือเล่มเดียวเป็นเหมือนบทสนทนาที่ดำเนินต่อไปข้ามกาลเวลา

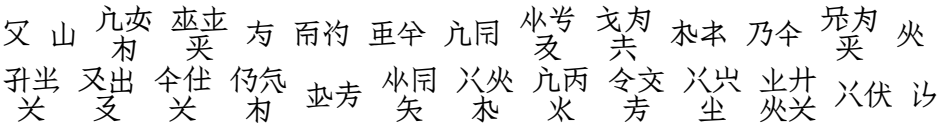
5.4 Egyptian Hieroglyphic Composition

The opening of the *Tale of the Shipwrecked Sailor*, lines 1–15 of Bob Richmond's horizontal UMdC transcription, is set as one continuous flowing Egyptian passage. The manuscript itself is in hieratic; the presentation below uses hieroglyphic transcription. Editorial line numbers and colour markup have been removed, while the original Unicode hieroglyphic signs and format controls remain.

Source: UMdC, ShipwreckedSailorHorizontal.umdc, opening lines 1–15; adapted from JSesh examples with reference to James P. Allen.

5.5 Khitan Small Script

The opening of the 1134 *Record of the Langjun*, an attested Khitan Small Script inscription, is transcribed in two-dimensional Unicode clusters. The first two lines of the Wikisource transcription form one flowing paragraph here; original transcription line breaks are omitted and cluster boundaries are preserved.



Source: Wikisource, 大金皇弟都統經略郎君行記, opening two transcription lines (Unicode transcription, CC BY-SA). This is a horizontal setting of a traditionally vertical inscription.

6 East Asian Writing and CJK Family Selection

This chapter compares document-level CJK typography with explicit local families for Chinese, Korean, Japanese, Vietnamese, and Tangut. The same Han codepoints may take different regional forms.

6.1 Chinese

Four paragraphs distinguish Simplified Chinese, mainland-style Traditional Chinese, Taiwan Traditional Chinese, and the document's Shanggu family. The three Traditional Chinese specimens use exactly the same source text.

Simplified Chinese — Noto SC 秋天的清晨，图书馆刚刚开门。书架上放着一本旧地图，画有山谷、城门与河流，旁边是《龙城纪事》。读书的人翻开书页，读到国王命令学者修订典籍的故事：有人研究龙骨，也有人制作木具，记下直角的形状。窗外下起雨，墨色在灯下格外清晰。

Traditional Chinese — Noto SC 秋天的清晨，圖書館剛剛開門。書架上放著一本舊地圖，畫有山谷、城門與河流，旁邊是《龍城紀事》。讀書的人翻開書頁，讀到國王命令學者修訂典籍的故事：有人研究龍骨，也有人製作木具，記下直角的形狀。窗外下起雨，墨色在燈下格外清晰。

Traditional Chinese — Noto TC 秋天的清晨，圖書館剛剛開門。書架上放著一本舊地圖，畫有山谷、城門與河流，旁邊是《龍城紀事》。讀書的人翻開書頁，讀到國王命令學者修訂典籍的故事：有人研究龍骨，也有人製作木具，記下直角的形狀。窗外下起雨，墨色在燈下格外清晰。

Traditional Chinese — Shanggu (document default) 秋天的淸晨，圖書館剛剛開門。書架上放著一本舊地圖，畫有山谷、城門與河流，旁邊是《龍城紀事》。讀書的人翻開書頁，讀到國王命令學者修訂典籍的故事：有人研究龍骨，也有人製作木具，記下直角的形狀。窗外下起雨，墨色在燈下格外淸晰。

6.2 Korean and Historical Hangul

The first two passages compare Hangul-only prose with a mixed Hanja–Hangul register. Both use the Korean family.

Modern Korean 가을비가 그친 오후, 작은 도서관의 창가에는 햇빛이 조용히 들어왔다. 학생들은 오래된 지도를 펼쳐 놓고 사라진 마을의 이름을 찾아보았다. 한 사람은 책의 여백에 질문을 적었고, 다른 사람은 다음 주에 다시 와서 자료를 읽기로 했다.

Mixed Hanja and Hangul 古書를 정리하던 학생은 朝鮮時代의 日記 한 권을 발견했다. 紙面에는 한글과 漢字가 나란히 적혀 있었고, 여러 사람의 손으로 남긴 註釋도 보였다. 학생은 낡은 책을 조심스럽게 넘기며 당시의 生活과 言語를 상상했다.

Middle Korean — *Hunminjeongeum eonhae* This continuous excerpt from the *Hunminjeongeum eonhae* retains arae-a, archaic finals, and tone marks. Its Unicode input includes precomposed modern syllables followed by archaic final jamo: the local Hangul shaper must compose these just as it composes their canonically equivalent decomposed sequences.

나·랏:말ᄊᆞ미 中듕國·귁·에달·아 文문字·쭝·와·로서르ᄉᆞᄆᆞᆺ·디아·니ᄒᆞᆯ·씨 ·이런젼·ᄎᆞ·로어·린 百·빅姓·셩·이니르·고·져·홇·배이·셔·도

6.3 Japanese

Ordinary modern Japanese combines kanji, hiragana, katakana, punctuation, and regionally selected Han forms within the same prose paragraph.

春の朝、京都の小さな書店に新しい本が届いた。店主は箱を開け、「歴史と文化」の棚に本を並べた。古い地図には城門や学校が描かれ、表紙には龍の絵がある。来店した学生はカタログで目当ての一冊を探し、静かな店内でページをめくった。

6.4 Vietnamese

These parallel excerpts reproduce the opening of the 2013 Vietnamese Constitution's preamble. The same Quốc Ngữ text appears first in the document's Libertinus family and then in the dedicated Vietnamese family. The Hán-Nôm specimen is a modern transcription of the same passage, not a historical constitutional manuscript; the local regular font is Minh Nguyen.

Quốc Ngữ — Libertinus Trải qua mấy nghìn năm lịch sử, Nhân dân Việt Nam lao động cần cù, sáng tạo, đấu tranh anh dũng để dựng nước và giữ nước, đã hun đúc nên truyền thống yêu nước, đoàn kết, nhân nghĩa, kiên cường, bất khuất và xây dựng nên nền văn hiến Việt Nam. Từ năm 1930, dưới sự lãnh đạo của Đảng Cộng sản Việt Nam do Chủ tịch Hồ Chí Minh sáng lập và rèn luyện, Nhân dân ta tiến hành cuộc đấu tranh lâu dài, đầy gian khổ, hy sinh vì độc lập, tự do của dân tộc, vì hạnh phúc của Nhân dân. Cách mạng tháng Tám thành công, ngày 2 tháng 9 năm 1945, Chủ tịch Hồ Chí Minh đọc Tuyên ngôn độc lập, khai sinh ra nước Việt Nam dân chủ cộng hòa, nay là Cộng hòa xã hội chủ nghĩa Việt Nam.

Quốc Ngữ — Vietnamese family Trải qua mấy nghìn năm lịch sử, Nhân dân Việt Nam lao động cần cù, sáng tạo, đấu tranh anh dũng để dựng nước và giữ nước, đã

hun đúc nên truyền thống yêu nước, đoàn kết, nhân nghĩa, kiên cường, bất khuất và xây dựng nên nền văn hiến Việt Nam. Từ năm 1930, dưới sự lãnh đạo của Đảng Cộng sản Việt Nam do Chủ tịch Hồ Chí Minh sáng lập và rèn luyện, Nhân dân ta tiến hành cuộc đấu tranh lâu dài, đầy gian khổ, hy sinh vì độc lập, tự do của dân tộc, vì hạnh phúc của Nhân dân. Cách mạng tháng Tám thành công, ngày 2 tháng 9 năm 1945, Chủ tịch Hồ Chí Minh đọc Tuyên ngôn độc lập, khai sinh ra nước Việt Nam dân chủ cộng hòa, nay là Cộng hòa xã hội chủ nghĩa Việt Nam.

Hán-Nôm — Minh Nguyen Regular 𨑗過尒𢆥𢆥歷史、人民越南勞動勤劬、創造、鬥爭英勇抵𨁪渃吧𡨸渃、㐌熏爈𢧚傳統愢渃、團結、仁義、堅強、不屈吧𡏦𨁪𢧚𡊲文獻越南。自𢆥1930、𨖅事領導貼黨共產越南由主席胡志明創立吧𤈜鍊、人民些進行局鬥爭數𨱽、㵋艱苦、犧牲爲獨立、自由貼民族、爲幸福貼人民。革命𦝄𠚢成功、𣈜2𦝄9𢆥1945、主席胡志明讀宣言獨立、開生𡃢渃越南民主共和、𦓡羅共和社會主義越南。

6.5 Tangut

The following paragraph restores the original Unicode Tangut corpus specimen in full, including its editorial punctuation. The text is a composite of short attested sequences from different Tangut works on Wikisource, not a single continuous historical document.

[illegible]，[illegible]，[illegible]，[illegible]，[illegible]，[illegible]，[illegible]，[illegible]，[illegible]，[illegible]，[illegible]，[illegible]，[illegible]，[illegible]，[illegible]。

6.6 WenJin Multi-Plane Coverage

A single local WenJin family covers CJK characters in the BMP and in supplementary planes through its registered P0, P2, and P3 font fallbacks. Table 4 supplies a fixed Unicode coverage matrix alongside the mixed-plane prose specimen.

古籍中的漢字有時仍未收入基本多文種平面。一位研究者翻檢舊鈔本，發現卷末記載的城門、山川、村社與古人名，當中出現了罕用的乙、吉、圡、堅、不。另一本字書則收錄丅、丰、乇、冂、巨等第三平面的文字。這些字在典籍裏並不是一串孤立的編碼，而是與普通漢字共同構成完整的語句。編輯將不同寫本的異體並列於校記，讀者則沿著正文閱讀，不必知道每個字屬於哪個 Unicode 平面，也不必在段落中手動切換字體。

6.7 Regional Forms of Shared Han Characters

Table 5 compares the same Unicode codepoints across five regional families; visible differences come from font selection or fallback, rather than from substituting different encoded characters.

7 Document Features

IMPE's feature system prepares selected, composable LaTeX package stacks. The specimens below use native package interfaces rather than the historical convenience wrappers.

7.1 Mathematics

The `math` feature supplies `amsmath` and `mathtools`. These independent identities test matrices, integrals, fractions, accents, cases, scripts, and mathematical alphabets without presenting unre-

Table 4: WenJin CJK block coverage (twenty codepoints per block)

Unicode block and range	Specimen
CJK Unified Ideographs U+4E00--U+9FFF	一丁丂七丄丅丆万丈三上下丌不与丐丑丒专
Extension A U+3400--U+4DBF	丕丙丠个丰乂乃乞乩乭乺乾亂亝亞亟亾亹亿
Extension B U+20000--U+2A6DF	亡亏亿亼亾亏仡且丘丘丣丈丌丗丛丯丳引
Extension C U+2A700--U+2B73F	廾节瓦隼奇奡猡臉氐帡冫夷屆乖秖狩邳枣倢垒
Extension D U+2B740--U+2B81F	丸乖亦爾久灰今侖侮倉傳僜奘岡凅渗前勢卑悊
Extension E U+2B820--U+2CEAF	七不卡东耒匝所轩皂兹趴尧秬杯豽豝戏昱番碖
Extension F U+2CEB0--U+2EBEF	寸才几万刁号丐歹考彐号豕旭尧屯审芬玊卅丆
Extension I U+2EBF0--U+2EE5F	丌舟牛焚仉仿伀仅仿佩价佂侥俣倪俞傕復拿龰
Extension G U+30000--U+3134F	丅丰丘冂巨丸立丐专申屮巡吊砑垚垠冤豺鬲甄
Extension H U+31350--U+323AF	元去陕垚陇來卅浮劾邮爹呇郤郪荚骁皋稠碕篡
Extension J U+323B0--U+3347F	五北冂訐垦葙衅銚锴鄢禾贱卻棚轋灺乖卒吵忝

Table 5: Regional forms of shared Han codepoints

Character	Shanggu	Noto SC	Noto TC	Japanese	Korean
骨	骨	骨	骨	骨	骨
直	直	直	直	直	直
令	令	令	令	令	令
具	具	具	具	具	具
角	角	角	角	角	角
門	門	門	門	門	門
國	國	國	國	國	國
學	學	學	學	學	學
龍	龍	龍	龍	龍	龍

lated formulae as a single derivation.

For a positive-definite matrix, a Gaussian integral evaluates to

$$A = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}, \qquad \det A = 4, \quad \beta > 0,$$

$$\begin{aligned} \mathcal{Z}(\beta) &= \int_{\mathbb{R}^3} \exp\!\left(-\frac{\beta}{2}\mathbf{x}^{\mathsf{T}} A \mathbf{x}\right) \mathrm{d}^3\mathbf{x} \\ &= \left(\frac{2\pi}{\beta}\right)^{3/2} \frac{1}{\sqrt{\det A}} = \frac{1}{2}\left(\frac{2\pi}{\beta}\right)^{3/2}. \end{aligned}$$

For $k \in \mathbb{N}_0$, the residue theorem provides a series and a contour integral:

$$f_k(z) = \frac{e^z}{z^{k+1}} = \sum_{n=0}^{\infty} \frac{z^{n-k-1}}{n!},$$

$$\frac{1}{2\pi i}\oint_{|z|=1} f_k(z)\,\mathrm{d}z = \operatorname{Res}_{z=0} f_k(z) = \frac{1}{k!}.$$

For integrable f, its Fourier transform illustrates wide accents:

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i x \xi}\,\mathrm{d}x, \qquad \overline{\widehat{f}(\xi)} = \widehat{\bar{f}}(-\xi).$$

A smooth flat function demonstrates a cases environment:

$$g(x) = \begin{cases} e^{-1/x^2}, & x > 0, \\ 0, & x \le 0, \end{cases} \qquad g^{(r)}(0) = 0 \quad (r = 0, 1, 2, \ldots).$$

Chebyshev's inequality and the decomposition of exterior powers supply independent examples of further mathematical notation:

$$\underbrace{\mathbb{P}(|X - \mathbb{E}[X]| \ge t)}_{\text{tail probability}} \le \frac{\operatorname{Var}(X)}{t^2}, \qquad t > 0.$$

$$\bigwedge^{k}(V \oplus W) \cong \bigoplus_{p+q=k} \left(\bigwedge^{p} V\right) \otimes \left(\bigwedge^{q} W\right).$$

A short inline comparison shows how the same summation and fraction occupy different amounts of vertical space:

Default display style $\displaystyle\sum_{j=1}^{n} \frac{1}{j^2}$

Local text style $\sum_{j=1}^{n} \frac{1}{j^2}$

7.2 Drawing

Figure 1 shows a Gaussian function with its formula and graduated axes. Figure 2 places a schematic manuscript-copying tree within the page.

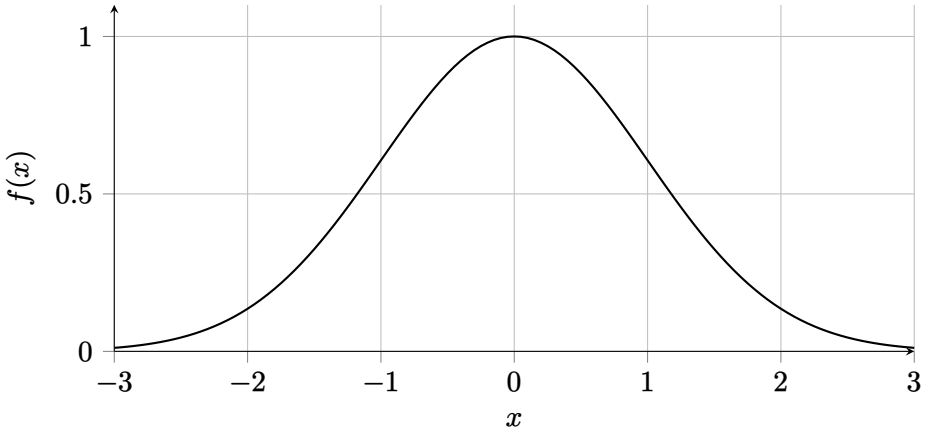


Figure 1: The Gaussian profile $f(x) = e^{-x^2/2}$.

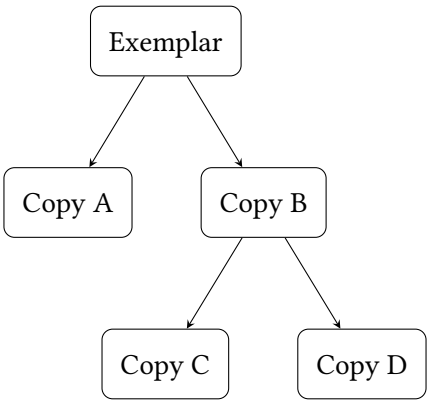


Figure 2: A schematic manuscript-copying tree, not a historical stemma.

7.3 Tables

Table 6 uses the native `table`, `tabularx`, and `booktabs` interfaces loaded by the `tables` feature.

Table 6: Feature integration summary

Feature	Public route	Exercised here
Mathematics	`math`	yes
Tables	`tables`	yes
Images	`image`	yes
Inline glyphs	`glyphs`	yes
Drawing	`drawing`	yes
Citations	`citations`	yes
Index	`index`	yes
Hyperlinks	`hyperlinks`	yes
Running headers	`headers`	yes

7.4 Lists and Quoted Examples

Ordinary lists and quotations do not require an IMPE-specific environment:

1. automatic font ownership;
2. script-specific behavior where required;
3. reusable document features;
4. stable public interfaces.

The native `quote` environment supplies an indented specimen:

> *A canonical showcase should be readable as a document, useful as a visual regression specimen, and representative of the public interface that new documents are expected to use.*

7.5 Images

The `image` feature prepares standard graphics and caption packages. Figure 3 includes a bundled vector image without relying on optional example-image files from the user's TeX distribution. Figure 3 also exercises native cross-references.

7.6 Inline Image Glyphs

A short passage from Chu bamboo manuscripts sets individual manuscript signs directly within the running text. The images function as characters, not floating figures; the transcriptions and source references are reproduced without adopting any particular decipherment.

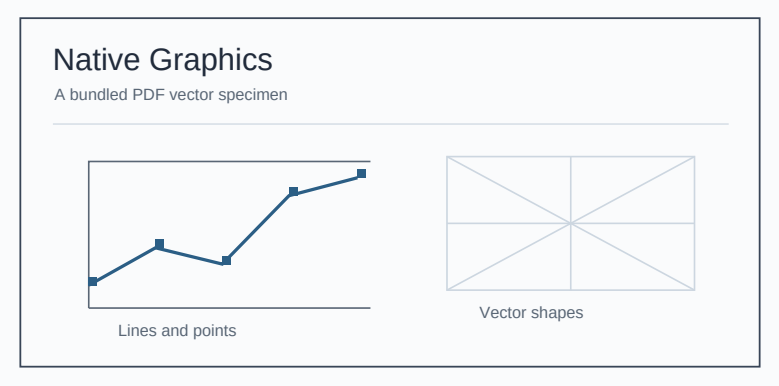


Figure 3: Native graphics inclusion using a bundled vector specimen.

「忮」字的古文字字形爲「 」（《郭店簡・語叢四》，15.14）、「 」（《上博簡・曹沫之陳》，30.16），一般隸定爲「忮」或「攲」。在出土楚地簡帛文獻中，本字的出現如下：

1 凡敓之道，級者爲首。既得其級，言必又及$_{=五}$之而弗亞，必隶其古。隶之而悘，必忮鉿$_{=}$。$_{十五}$之$_{=}$而不可，必𠬜以訛，毋命智我。皮邦芒$_{六}$牏，流澤而行。$_{七}$（《郭店簡・語叢四》）[2]

2 ……□厚飤，思爲前行。三行之逡，句見耑兵，忮$_{三十}$（《上博簡・曹沫之陳》）[3]

「鉿」的古文字字形爲「 」，在出土楚系簡帛文獻中僅見於上述文段 1。

7.7 Citations and Bibliography

Citation support is loaded through the feature system and configured before feature activation. A normal textual citation can therefore be written as Knuth (1984), while a parenthetical citation appears as (Knuth, 1984).

The bibliography is printed near the end of the document.

7.8 Index Terms

Earlier sections use the native `\index` command to register concepts at their actual occurrences. The phrase *canonical showcase* is indexed here; native interfaces also provide a hierarchical example. IMPE supplies unique occurrence-level destinations for its default index, while `imakeidx` and `xindy` retain their usual indexing syntax.

7.9 Hyperlinks, Footnotes, and Running Headers

The table of contents is clickable, numbered headings carry stable targets, and section headings can link back toward their table-of-contents entries. The running header above this page is produced by the `headers` feature.

Footnote markers are also bidirectional.[4]

For an external-link specimen, the project repository can be represented with a normal hyperlink: IMPE LaTeX System on GitHub.

[2] 武漢大學簡帛研究中心、荊門市博物館編著：《楚地出土戰國簡冊合集（一）：郭店楚墓竹書》，文物出版社，2011 年，167–168 頁。

[3] 馬承源主編：《上海博物館藏戰國楚竹書（四）》，上海古籍出版社，2004 年，262 頁。

[4] Clicking the footnote marker jumps to this note; the note marker can link back to the original position.

References

Knuth, D. E. (1984). *The texbook*. Addison-Wesley.